

BRAC University
CSE331 : Automata And Computability
Spring 2025
Quiz 3
SET - A

Marks :20

Duration: 30 mins

Name:	ID:	Sec:
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1. Using Pumping Lemma, prove that the following languages are not regular.

a. $L = \{ w \in \{a,b,c\}^* : w = a^i b^j c^k, \text{ where } i+j > k \text{ and } i,j,k \geq 0 \}$ [6]

2. Take a look at the grammar below and solve the following problems.

$$\begin{aligned} S &\rightarrow AX \\ A &\rightarrow 0A \mid 1B \mid \epsilon \\ B &\rightarrow 0B \mid 1A \\ X &\rightarrow 00X \mid 01X \mid 10X \mid 11X \mid 0 \mid 1 \end{aligned}$$

- (a) Give a leftmost derivation for the string 01101010. [4]
- (b) Sketch the parse tree corresponding to the derivation you gave in (a). [4]
- (c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. [5]
- (d) Find a string w of length six such that w has exactly one parse tree in the grammar above. [1]

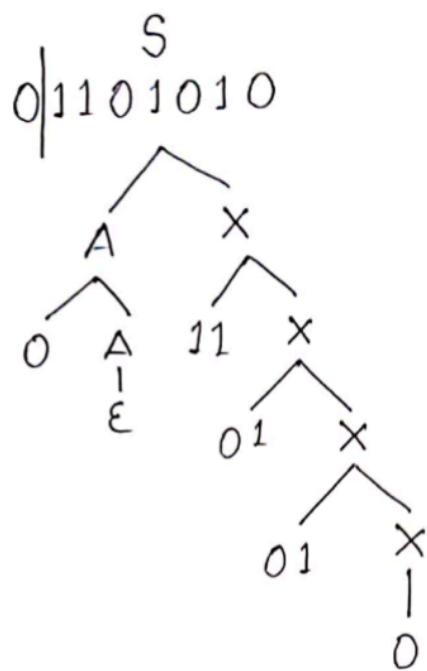
1. Let,
 L is regular language,
 P is the pumping length. let's
 take a string w of pumping
 length P .

$$w = a^{\frac{P}{2}} b^{\frac{P}{2}} c^{\frac{P}{2}} ; |w| = \frac{3P}{2} > P$$

$$i = 1; \quad \underbrace{a^{\frac{P}{2}} b^{\frac{P}{2}}}_y \underbrace{c^{\frac{P}{2}}}_z$$

$$\text{now, } i = 2; \quad \frac{1}{x} \underbrace{a^{\frac{P}{2}} b^{\frac{P}{2}}}_y \underbrace{a^{\frac{P}{2}} b^{\frac{P}{2}}}_y \underbrace{c^{\frac{P}{2}}}_z$$

after pumping the string
 it breaks the sequence so
 language is not regular.



S
 $\rightarrow AX$
 $\rightarrow 0AX$
 $\rightarrow 0\epsilon X$
 $\rightarrow 011X$
 $\rightarrow 01101X$
 $\rightarrow \underline{0110101X}$
 $\rightarrow 01101010$

